Lab 2.1

Reviewing Requirements Documents

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Exercise 1 TTPU library system

1.1 Contradictions

Pair	Why they contradict
FR-01 and FR-02	The maximum is 5 books, but FR-02 permits a 6th book.
FR-04 and FR-06	A reference book may not leave the building, yet FR-06 treats it as an ordinary renewable loan.
FR-07 and FR-08	The same reminder is required both 2 days before and only after the due date.
FR-16 and FR-17	Full history forever conflicts with deletion of personal data 1 year after departure.

1.2 Lines that are not requirements

ID	What it really is	Where it belongs
FR-13	Design decision: MySQL 8 is an implementation technology.	Design document after technical evaluation.
FR-18	Business/organisational goal: purchasing books is not system behaviour.	Library business plan or project plan.
NFR-03	Impossible promise: no software can be guaranteed never to fail.	Nowhere; replace it with a measurable reliability target.

1.3 Untestable lines repaired

ID	Corrected requirement	Test
FR-09	NFR-04 A first-year student shall find and reserve a known available book in no more than 4 clicks after sign-in.	Ask 5 first-year students to reserve a named book and count clicks.
FR-14	NFR-05 Search results shall appear within 2 seconds for a catalogue of 50,000 books and 100 concurrent users.	Run 100 searches against 50,000 records and measure response time.
FR-15	NFR-06 The first page of search results shall appear within 2 seconds for 95% of 100 test searches.	Execute 100 representative searches and time each result.
NFR-03	NFR-03 The system shall have at least 99.5% availability during library working hours each semester.	Compare monitoring downtime with scheduled working hours.

1.4 Contradictions repaired

• FR-01 The system shall allow a registered student to borrow at most 5 books without special approval.

• FR-02 The system shall allow a librarian to approve one additional book, increasing the limit to 6 for that loan.

• FR-04 Books marked Reference shall be used only inside the library and shall not create a take-home loan.

• FR-06 The system shall not offer online renewal for a book marked Reference.

• FR-07 The system shall send an email reminder 2 days before the due date for a loan that is not returned.

• FR-08 The system shall send one overdue email at 09:00 on the first day after the due date if the book remains unreturned.

• FR-16 The system shall retain borrowing transactions in anonymised form indefinitely for statistics.

• FR-17 The system shall delete or anonymise data identifying a student within 1 year after the student leaves the university.

Exercise 2 TTPU canteen pre-order system

2.1 Full review table

ID	Defect	What is wrong	Corrected line or action
FR-03	C	Contradicts FR-02 on when and how payment occurs.	The student shall pay in the application; cash payment is not supported in version 1.
FR-06	C	Changes until 11:45 conflict with cooking starting at 11:30.	A student shall change an order until 11:25 while its status is Not started.
FR-08	C	Contradicts FR-09 about zero-stock dishes.	Delete; FR-09 remains the stock rule.
FR-11	C	Immediate delivery conflicts with the single list at 11:00.	Each confirmed order shall appear on the kitchen screen within 5 seconds.
FR-12	C	Conflicts with FR-11 and can delay the kitchen.	At 11:00 the system shall show a summary of all confirmed orders in addition to individual orders.
FR-13	A/T/W	Work without internet is ambiguous, unmeasurable and is non-functional in the functional list.	NFR-05 The kitchen screen shall show cached confirmed orders for up to 30 minutes after internet loss.
FR-15	C	English-only messages conflict with three interface languages.	Messages shall use the language selected by the user from Uzbek, Russian and English.
FR-17	N	Django is a design decision, not required behaviour.	Delete and evaluate technology in the design document.
FR-18	N	Reducing waste is a business goal, not software behaviour.	Move to project success criteria; use reports to measure it.
FR-19	T/W	Secure is unmeasurable and is non-functional.	NFR-04 Payment data shall use TLS 1.3 and no CVV shall be stored.
FR-20	N	A delivery deadline is a project constraint, not system behaviour.	Move to the project plan as a milestone.
NFR-03	T	Easy to use is subjective and has no measure.	NFR-03 A first-year student shall place a standard order in at most 6 clicks without training.

2.2 Missing requirements

• FR-21 The system shall mark a paid order Uncollected if it is not collected within 60 minutes after it is marked Ready.

• FR-22 The system shall return a cancelled card payment to the original payment method within 5 working days.

• FR-23 The system shall show declared allergens for every dish before the student confirms the order.

• FR-24 The system shall prevent ordering on dates marked Closed in the manager calendar.

• FR-25 Only an authenticated user with the Manager role shall open the manager screen and view revenue.

2.3 SDLC timing and late cost

ID	Found now	Should be found	If found after release
FR-03	Requirements review	Requirement gathering/review	About 50-100 units: payment flow, UI, accounting and training would need correction.
FR-06	Requirements review	Requirement review	About 50 units: students could change food already cooking, causing refunds and waste.
FR-11	Requirements review	Requirement review	About 50-100 units: the kitchen screen and staff workflow would need redesign and retraining.
FR-15	Requirements review	Requirement review/prototyping	About 50 units: messages and translations across all screens would need rework after users complain.

2.4 Use case Student orders a meal

Actor: Student. Precondition: the student is signed in, ordering is open from 08:00 to 11:00, and at least one dish has portions available.

• 1. The system shows today's menu, prices, available portions and allergens.

• 2. The student selects one or more available dishes.

• 3. The system shows the order summary and total amount.

• 4. The student confirms the order and chooses application payment.

• 5. The payment service confirms payment.

• 6. The system assigns a daily unique order number.

• 7. The order appears on the kitchen screen within 5 seconds.

• 8. The system shows the order number and confirmation to the student.

Alternative flow - payment fails: At step 5, if payment is rejected or times out, the system does not create a confirmed kitchen order, shows the failure reason, and lets the student retry once or cancel.

Postcondition: a paid order with a unique number is stored and visible to the kitchen, or no confirmed order exists if payment failed. Traceability: FR-01, FR-02, FR-04, FR-07, FR-09, FR-10, corrected FR-11, FR-21, FR-23 and NFR-01.